

Blind image watermarking method based on linear canonical wavelet transform and QR decomposition

Yong Guo^{1,2}, Bing-Zhao Li^{1,2} ✉

¹School of Mathematics and Statistics, Beijing Institute of Technology, Beijing 100081, People's Republic of China

²Beijing Key Laboratory on MCAACI, Beijing Institute of Technology, Beijing 100081, People's Republic of China

✉ E-mail: li_bingzhao@bit.edu.cn

ISSN 1751-9659

Received on 10th December 2015

Revised 6th May 2016

Accepted on 16th May 2016

doi: 10.1049/iet-pr.2015.0818

www.ietdl.org

Abstract: Inspired by the fact that wavelet transform can be written as a classical convolution form, a new linear canonical wavelet transform (LCWT) based on generalised convolution theorem associated with linear canonical transform (LCT) is proposed recently. The LCWT not only inherits the advantages of multi-resolution analysis of wavelet transform (WT), but also has the capability of image representations in the LCT domain. Based on these good properties, the authors propose a novel image watermarking method using LCWT and QR decomposition. Compared with the existing image watermarking methods based on discrete WT or QR, this novel image watermarking method provides more flexibility in the image watermarking. Peak signal-to-noise ratio, normalised cross and structural similarity index measure are used to verify the advantages of the proposed method in simulation experiments. The experiment results show that the proposed method is not only feasible, but also robust to some geometry attacks and image processing attacks.

1 Introduction

Nowadays, digital image can be widely spread all over the world by internet. Therefore, the protection of image becomes a challenge problem. In order to overcome the problem, many kinds of information hiding technologies are proposed, for example steganography, digital watermarking and visual cryptography. As one of the important branches of information hiding technologies, the digital image watermark technology is a digital protection technology for the purpose of secret transmission, storage and identification. The digital watermark technology has become one of the hottest research topics in information security community [1–6].

Feasibility and robustness are the main requirements for developing a new digital image watermarking method [7]. The digital image watermarking method can be classified into spatial domain method and transform domain method. The transform domain digital watermarking method modifies the transform coefficients to embed watermark, which is more robust against many image processing attacks in comparison to the spatial domain method. Currently, the discrete cosine transform (DCT), the discrete Fourier transform (FT) and the discrete wavelet transform (DWT) are important transforms in transform domain digital watermarking method, where DWT has attracted widespread attentions. Some image watermarking methods based on the transforms associated with DWT are proposed recently, such as lifted wavelet transform, integer wavelet transform and redundant wavelet transform [8–10]. Due to its advantages like multi-resolution representation and better energy compression, the image watermarking method based on DWT can provide good robustness and feasibility. It has been shown in [11] that some watermarking methods using DWT alone show good robustness to image processing attacks, but bad robustness to some geometric attacks. In order to improve the robustness to common geometric attacks, the image watermarking methods based on matrix factorisation are proposed recently, such as singular value decomposition (SVD) and QR decomposition [12–17]. Moreover, it has been proved that the image watermarking method based on QR decomposition offers better or at least comparable performance than SVD in some metrics [17]. The digital image watermarking method based on DWT and matrix decomposition combines the advantages of DWT and matrix decomposition, which is becoming a hot topic in image watermarking field [18–20].

The linear canonical transform (LCT) is a generalisation of FT and has been proved to be a powerful tool in signal processing and time–frequency analysis [21–26]. However, for the best of our knowledge, there are no research papers published associated with the application of a combination of WT and LCT. It is therefore interesting and worthwhile to investigate the applications of the combination of WT and LCT.

In this paper, a new kind of transform named as linear canonical wavelet transform (LCWT) by combining the WT and LCT is introduced [27]. Moreover, like WT and LCT, LCWT has potential applications in the area of image processing, such as image fusion, image watermarking and image steganography and so on. Moreover, a novel image watermarking method is derived by the newly introduced transform and QR decomposition. The remaining of this paper is organised as follows. In Section 2, we review some basic concepts about LCT, the definition and basic properties of LCWT. In Section 3, a novel image watermarking method is proposed based on the newly introduced LCWT and QR decomposition. The performance analysis of the novel image watermarking method is shown in Section 4. Conclusions and future works are stated in Section 5.

2 Linear canonical wavelet transform

2.1 Linear canonical transform

The LCT with real parameter matrix A_0 of a signal $f(t)$ is defined as [21]

$$F_{A_0}(u) = \int_{\mathbb{R}} f(t) K_{A_0}(t, u) dt \quad (1)$$

with the kernel

$$K_{A_0}(t, u) = \begin{cases} \frac{1}{\sqrt{ib_0}} \exp\left(i \frac{a_0 t^2 + d_0 u^2 - 2tu}{2b_0}\right), & b_0 \neq 0, \\ \sqrt{d_0} \exp\left(i \frac{c_0 d_0 u^2}{2}\right) \delta(u - d_0 t), & b_0 = 0, \end{cases} \quad (2)$$

where

$$A_0 = \begin{pmatrix} a_0 & b_0 \\ c_0 & d_0 \end{pmatrix}$$

satisfying $a_0 d_0 - b_0 c_0 = 1$ and $F_{A_0}(u)$ is the LCT of $f(t)$.

At present, some well-known concepts and theorems associated with the classical FT, such as uncertainty principle, sampling theorem and the convolution theorem, have been derived in the LCT domain [21–26]. Shi *et al.* [25] introduced a canonical convolution with matrix parameter

$$A_1 = \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}$$

of $f(t), g(t) \in L^2(R)$ as follows:

$$f(t) \Theta_{A_1} g(t) = \int_{-\infty}^{+\infty} g(\tau) T_{\tau}^{A_1} f(t) d\tau \quad (3)$$

where Θ_{A_1} denotes the general canonical convolution operator, $T_{\tau}^{A_1} f(t) = f(t - \tau) e^{-i(a_1/b_1)\tau(t - \tau/2)}$. Also we can obtain

$$F_{A_1}[f(t) \Theta_{A_1} g(t)](u) = \sqrt{2\pi a} F_{A_1}(u) G(u/b_1) \quad (4)$$

where $F_{A_1}(u)$ indicates the LCT of $f(t)$, $G(u/b_1)$ denotes the FT of $g(t)$.

2.2 Definition of LCWT

It is well known that WT can be written as a classical convolution form as follows: (see (5)) where $a \in R^+$, $b \in R$ and $\psi_{a,b}(t)$ is a continuous affine transformation of the mother wavelet $\psi(t)$, i.e. $\psi_{a,b}(t) = (1/\sqrt{a})\psi((t-b)/a)$. $*$ denotes the classical convolution operator and $\langle \cdot, \cdot \rangle$ indicates the inner product in $L^2(R)$.

Based on the above facts, a new LCWT based on generalised convolution theorem associated with LCT is proposed in [27]. The one-dimension LCWT (1D-LCWT) of a signal $f(t)$ with matrix parameter A_1 is defined as following [27]:

$$\begin{aligned} W_f^{A_1}(a, b) &= f(t) \Theta_{A_1} \left(\frac{1}{\sqrt{a}} \psi \left(-\frac{t}{a} \right) \right) \\ &= e^{-i(b^2/2) \cdot (a_1/b_1)} \left\langle f(t) e^{i(t^2/2) \cdot (a_1/b_1)}, \psi_{a,b}(t) \right\rangle \\ &= e^{-i(b^2/2) \cdot (a_1/b_1)} \int_{-\infty}^{+\infty} (f(t) e^{i(t^2/2) \cdot (a_1/b_1)}) \psi_{a,b}(t) dt \end{aligned} \quad (6)$$

where matrix A_1 satisfying $a_1 d_1 - b_1 c_1 = 1$.

In further, we expand 1D-LCWT into 2D LCWT (2D-LCWT). The definition of 2D-LCWT is shown in following form: (see equation below) where

$$A_m = \begin{pmatrix} a_m & b_m \\ c_m & d_m \end{pmatrix} \quad (m = 1, 2).$$

Follow from (4) and the LCT of (6), we can obtain

$$F_{A_1}[W_f^{A_1}(a, b)] = F_{A_1} \left[f(t) \Theta_{A_1} \left(\frac{1}{\sqrt{a}} \psi \left(-\frac{t}{a} \right) \right) \right] = \sqrt{2\pi a} F_{A_1}(u) \Psi(u/b_1) \quad (7)$$

where $F_{A_1}(u)$ indicates the LCT of $f(t)$ and $\Psi(u/b_1)$ denotes the FT of $\psi(t)$.

So the LCWT can be expressed in terms of the LCT of signal $f(t)$ as follows:

$$W_f^{A_1}(a, b) = \int_{-\infty}^{+\infty} \sqrt{2\pi a} F_{A_1}(u) \Psi(u/b_1) K_{A_1}^*(b, u) du \quad (8)$$

whenever $a_1 = 0$, LCWT reduces to the classical WT, and if $a_1 = \cos \theta$, $b_1 = \sin \theta$, LCWT reduces to fractional wavelet transform proposed by Shi *et al.* [28].

2.3 Basic properties of LCWT

In this subsection, some basic properties of LCWT are listed and the detailed proof can be found in [27].

- Linear property:** If $f(t) = k_1 f_1(t) + k_2 f_2(t)$ and $f_1(t) \rightarrow W_{f_1}^{A_1}(a, b)$, $f_2(t) \rightarrow W_{f_2}^{A_1}(a, b)$, then it can be verified that LCWT satisfy linear property, i.e.

$$W_f^{A_1}(a, b) = k_1 W_{f_1}^{A_1}(a, b) + k_2 W_{f_2}^{A_1}(a, b) \quad (9)$$

- Scaling property:** If $f(t) \rightarrow W_f^{A_1}(a, b)$, $f(ct) = f(ct)$, then

$$W_{f(ct)}^{A_1}(a, b) = \frac{1}{\sqrt{c}} W_f^{A_1}(ca, cb) \quad (10)$$

where $c > 0$ and

$$A'_1 = \begin{pmatrix} a'_1 & b'_1 \\ c'_1 & d'_1 \end{pmatrix}$$

satisfying $(a'_1/b'_1) = (a_1/b_1) \cdot (1/c^2)$

- Inner product theorem:** Let $W_f^{A_1}(a, b)$ and $W_g^{A_1}(a, b)$ denote the LCWT of $f(t)$, $g(t)$, and $\Psi(\Omega)$ denotes the FT of $\psi(t)$. If $\Psi(\Omega)$ satisfies

$$C_{\Psi} = \int_{-\infty}^{+\infty} \frac{|\Psi(\Omega)|^2}{\Omega} d\Omega < \infty. \quad (11)$$

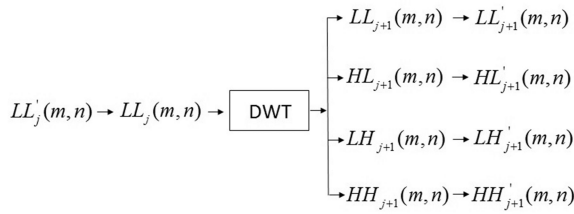
Then

$$\int_{R^2} W_f^{A_1}(a, b) [W_g^{A_1}(a, b)]^* \frac{1}{a^2} da db = 2\pi C_{\Psi} \int_{-\infty}^{+\infty} f(t) g^*(t) dt. \quad (12)$$

- Parseval's relation:** Let $f(t) = g(t)$, then we obtain Parseval's relation associated with LCWT as following:

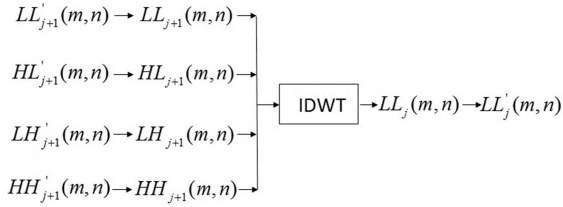
$$W_f(a, b) = \int_{-\infty}^{+\infty} f(t) \psi_{a,b}^*(t) dt = \langle f(t), \psi_{a,b}(t) \rangle = f(t) * \left(\frac{1}{\sqrt{a}} \psi \left(-\frac{t}{a} \right) \right) \quad (5)$$

$$\begin{aligned} W_f^{A_1, A_2}(a, k_1, k_2) &= e^{-i(1/2)(k_1^2 \cdot (a_1/b_1) + k_2^2 \cdot (a_2/b_2))} \cdot \int_{R^2} (f(x, y) e^{i(1/2)(x^2 \cdot (a_1/b_1) + y^2 \cdot (a_2/b_2))} \\ &\quad \cdot \frac{1}{a} \psi \left(\frac{x - k_1}{a} \right) \psi \left(\frac{y - k_2}{a} \right) dx dy \end{aligned}$$



The decomposition of LCWT

a



The construction of LCWT

b

Fig. 1 Discrete LCWT

(a) Decomposition of LCWT, (b) Construction of LCWT

$$\int_{\mathbb{R}^2} |W_f^A(a, b)|^2 \frac{1}{a^2} da db = 2\pi C_\psi \int_{-\infty}^{+\infty} |f(t)|^2 dt. \quad (13)$$

3 A novel blind digital image watermarking method

In this section, a novel digital image watermark method based on LCWT and QR decomposition is proposed. First the algorithm of discrete LCWT and QR decomposition is introduced. Then watermarking embedding and extraction procedures are shown in detail.

3.1 Discrete LCWT

The decomposition and construction of LCWT are introduced in this subsection. First, we will show the decomposition of LCWT. Similar to DWT, LCWT decompose image into four subbands: linear canonical domain low-frequency subband LL' , horizontal linear canonical domain high-frequency subband HL' , vertical linear canonical domain high-frequency subband LH' and linear canonical domain high-frequency subband HH' . Assume $LL'_j(m, n) (m, n \in \mathbb{Z}^+)$ is low-frequency subband coefficient of the j th level of linear canonical wavelet decomposition, which produced by $e^{(i/2)[(m \cdot 2^j)^2(a_1/b_1) + (n \cdot 2^j)^2(a_2/b_2)]}$ to obtain $LL_j(m, n)$, then followed by a WT to obtain low frequency coefficient $LL_{j+1}(m, n)$ and high frequency coefficient $HL_{j+1}(m, n)$, $LH_{j+1}(m, n)$ and $HH_{j+1}(m, n)$. Lastly, respectively, produced by $e^{-(i/2)[(2m \cdot 2^j)^2(a_1/b_1) + (2n \cdot 2^j)^2(a_2/b_2)]}$ to obtain $LL'_{j+1}(m, n)$,

$HL'_{j+1}(m, n)$, $LH'_{j+1}(m, n)$ and $HH'_{j+1}(m, n)$. The above procedure can be summarised in Fig. 1a. Moreover, the reconstruction of LCWT is the inverse process of decomposition, which is shown in Fig. 1b. Obviously, parameter matrices influence the energy distribution of low-frequency and high-frequency subbands, so following studying is focus on the parameter matrices. As shown in Fig. 2a, it can be seen from these images that the high frequency image include more information of original image with the increasing of α . The energy of high frequency becomes larger with the increasing of $\alpha = a_{11}/a_{12} = b_{11}/b_{12}$, which is shown in Fig. 2b.

3.2 QR decomposition

Any matrix A of size $m \times n$ can be factored as following:

$$A = QR \quad (14)$$

where Q is an $m \times n$ matrix satisfying $QQ^T = I$ and R is an $n \times n$ upper triangular matrix. If A is non-singular, this factorisation is unique.

It has been proved that the absolute value of elements of the first row of R matrix is larger than the absolute value of the other rows if the columns of A are correlated [17]. Based on the above fact, if A is an image matrix, small change in the first row of R matrix will not lead to image distortion. That is the reason why we embed watermark in the first row of R matrix in the following section. We will show the procedures of watermark embedding and watermark extraction. Watermark image is embedded into cover image to obtain watermarked image, which is used to extract watermark. Since original watermark image is not needed in watermark extraction, the proposed method is a blind digital watermark method.

3.3 Digital image watermark method

3.3.1 Watermark embedding: In this subsection, we will introduce our watermark embedding procedures. Assume the cover image $X_{512 \times 512}$ is a grey-level image. The watermark image $W_{64 \times 64}$ is a binary image of size 64×64 . The embedding procedures can be summarised in Fig. 3a and detailed elaborated as follows:

- The cover image X is converted into a column Y of size $262, 144 \times 1$.
- Y is resorted as increasing order to obtain Y' . Meanwhile record the original position of the element of Y' , it will be used in step 11.
- Y' is reshaped back to 512×512 image X' .
- Perform one level LCWT to X' to obtain four subbands LL' , HL' , LH' and HH' , where the linear canonical approximate subband LL' of size 256×256 will be used in the following steps.
- LL' is divided into non-overlapping blocks $\{b_{ij}\}_{4 \times 4}$ of size 4×4 , the number of all blocks are 64^2 .
- Apply QR decomposition to each block to obtain two components Q_{ij} and R_{ij} , where R_{ij} is the upper triangular matrix.
- The first row of each R_{ij} is selected to embed binary watermark as following:

$$R_{ij}^w(1, :) = \begin{cases} R_{ij}(1, :) + \lambda \cdot K & \text{if } w_{ij} = 1 \\ R_{ij}(1, :) - \lambda \cdot K & \text{if } w_{ij} = 0 \end{cases} \quad (15)$$

where K is an 1×4 -dimensional random integral vector, whose values are independent and uniformly distributed over the interval $[-1, 1]$. $R_{ij}(1, :)$ denotes the first row of R_{ij} , R_{ij}^w denotes the watermarked block. λ denotes the watermark embedding strength, which can be balance the invisibility and robustness in watermark method.

- Perform inverse QR to each block to obtain watermarked block $b_{ij}^w = Q_{ij}R_{ij}^w$.
- Merge all these watermarked blocks together to obtain $\{LL'\}^w$ as opposite to step 5, where $\{LL'\}^w$ is watermarked linear canonical approximate subband.
- Perform inverse LCWT to $\{LL'\}^w$ and other three subbands HL' , LH' and HH' obtained in step 4, watermarked scrambled cover image $\{X'\}^w$ can be obtained.
- Using the position vector in step 2, and we can obtain watermarked cover image X^w .

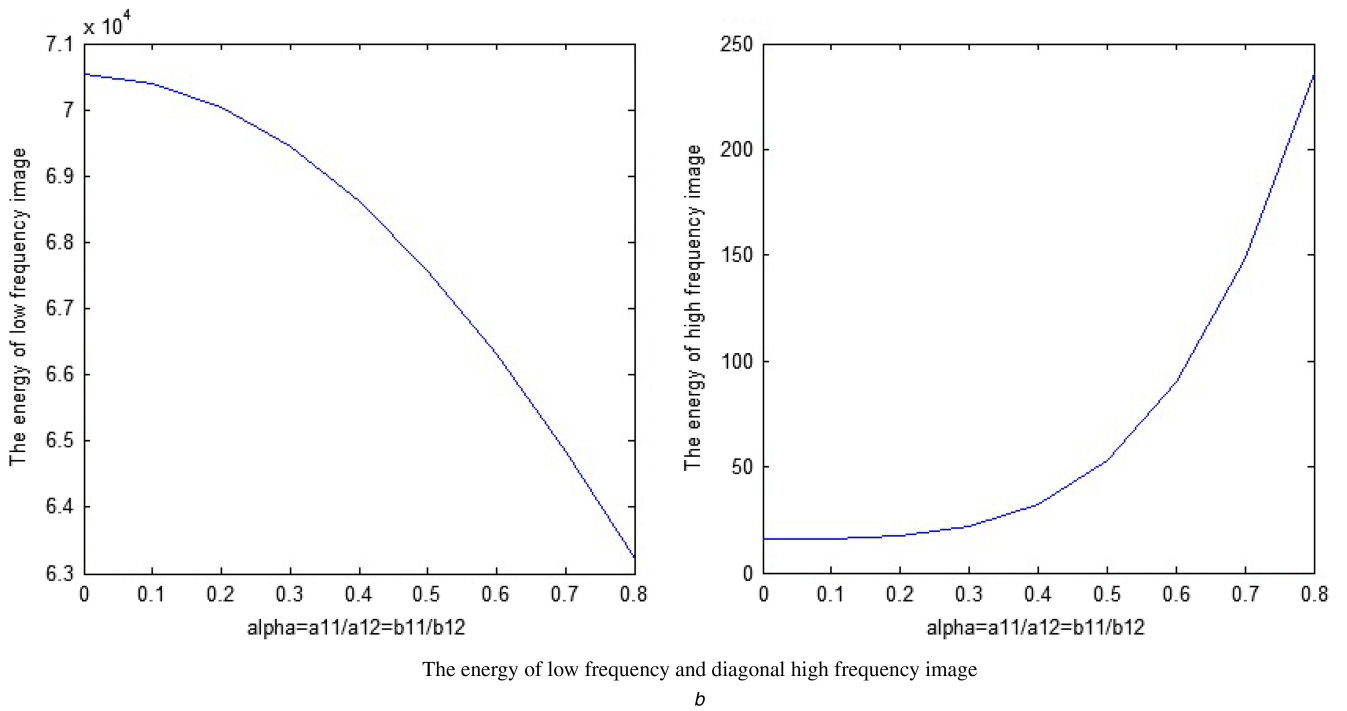
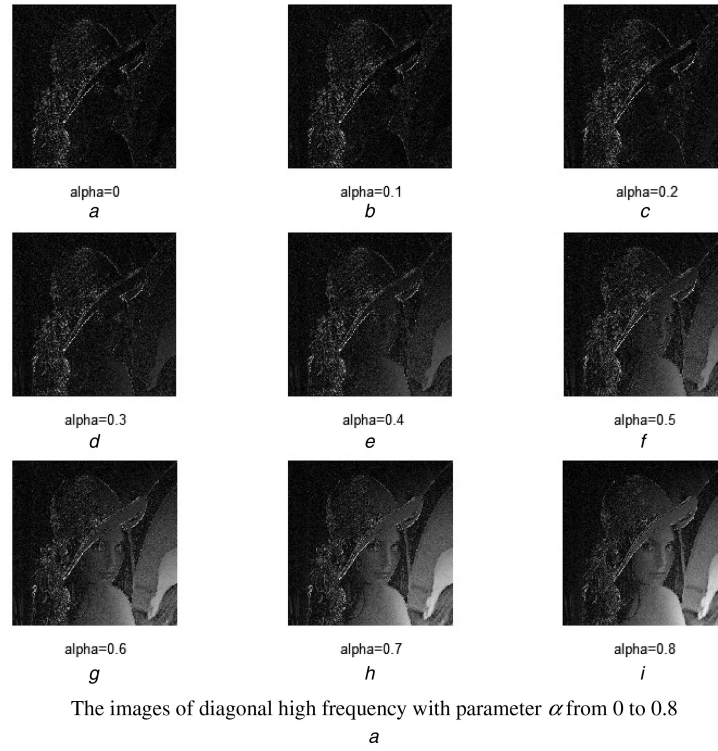


Fig. 2 Analysis of LCWT decomposition

(a) Images of diagonal high frequency with parameter α from 0 to 0.8, (b) Energy of low frequency and diagonal high frequency image

3.3.2 Watermark extraction: Next, we will show the watermark extraction scheme to extract binary watermark, which using K and the watermarked scrambled cover image $\{X\}^w$. The extraction procedure can be summarised in Fig. 3b and detailed presented as follows:

- i. Apply one level LCWT to the watermarked scrambled cover image $\{X\}^w$, we can obtain four subbands, where the watermarked linear canonical approximate subband LL1 of size 256×256 will be used in the following steps.
- ii. LL1 is divided into non-overlapping blocks $\{a_{ij}\}_{4 \times 4}$ of size 4×4 , the number of all blocks are 64^2 .

- iii. Apply QR decomposition to each 4×4 block a_{ij} to obtain two components $Q1_{ij}$ and $R1_{ij}$.
- iv. Watermark image is extracted from the first row of $R1_{ij}$ using (16) as follows:

$$w'(i, j) = \begin{cases} 1 & \text{if } \text{corrcoef}(R1_{ij}(1, :), K) \geq \text{corrcoef}(R1_{ij}(1, :), -K), \\ 0 & \text{if } \text{corrcoef}(R1_{ij}(1, :), K) < \text{corrcoef}(R1_{ij}(1, :), -K), \end{cases} \quad (16)$$

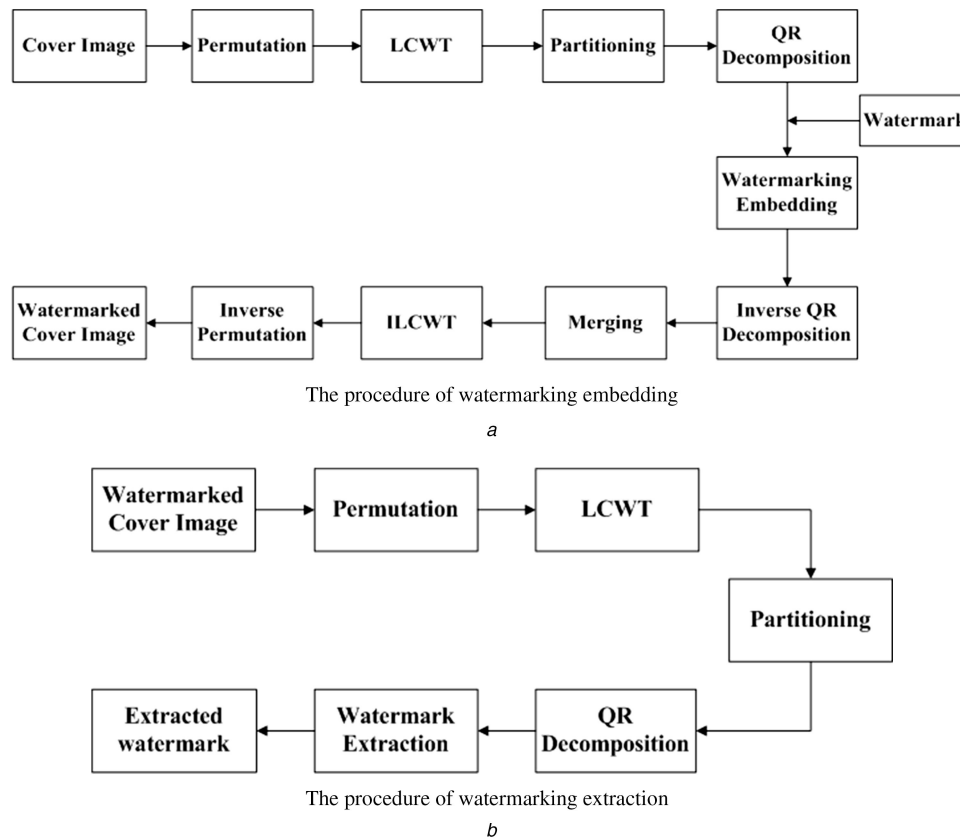


Fig. 3 Proposed watermarking method

(a) Procedure of watermarking embedding, (b) Procedure of watermarking extraction



Fig. 4 Test cover images

(a) Lena, (b) Boat, (c) Pepper, (d) Elaine, (e) Monkey, (f) Watermark

where $R1_{ij}(1, \cdot)$ denotes the first row of $R1_{ij}$. $\text{corrcoef}(\cdot, \cdot)$ is the standard correlation coefficient, which can be used to measure similarity.

- v. The technology is performed on extracted watermark to decrease the error, if the sum of eight pixel value around it are larger than 6, then $w'(i, j) = 1$, else if the sum of pixel value around it are less than 3, then $w'(i, j) = 0$. It is worth to point out that these threshold values are determined experimentally, and the above two threshold values can be changed according to practical need.

4 Simulation experiments

In this section, the feasibility and robustness of the proposed image watermark method are analysed by simulation experiments. All simulation experiments are conducted on a personal computer having Intel dual core 3.2 GHz CPU with 4.0 GB RAM, and using MATLAB version R2014a under the Windows 7 environment. Meanwhile, some normal images are used as the cover images, such as the Lena image, Elaine, Boat, Pepper and the Monkey images, which are shown in Figs. 4a–e, respectively. A 64×64 BIT image is used as watermark image, which is shown in Fig. 4f. The parameter settings of LCWT are $a_1/b_1 = a_2/b_2 = 0.05$, watermark strength is $\lambda = 15$. Here we take the watermark methods based on DWT and QR decomposition (abbreviated as DWT + QR) proposed in [18] and watermark method based on QR decomposition (abbreviated as QR) proposed in [17] as

comparison. Since the watermarking method based on DWT + QR and QR is popular and it has been shown that the QR decomposition offers capacity and robustness comparable to or better than a similar watermark method based on DCT and SVD [17]. Therefore, we re-implement the algorithm of QR and DWT + QR according to the procedures of watermark embedding and extraction mentioned in [17, 18] based on same experimental conditions. The corresponding results are also obtained in this way. In order to show the advantages of the proposed method, the follows are divided into two parts: the feasibility and robustness of the proposed image watermarking method. In the first part, watermark is embedded into five cover images and then extracted from the watermarked cover image. Peak signal-to-noise ratio (PSNR) is used to measure the watermark invisibility, normalised cross (NC) and structural similarity index measure (SSIM) are used to measure the similarity between the original watermark image and extracted watermark image. In the second part, we detect the watermark robustness to some geometry attacks (rotation, translation, cropping and scaling) and image processing attacks (JPEG compression, filter, addition of noise, Sobel sharpen, gamma correction, histogram equalisation and motion blur).

4.1 Feasibility of proposed method

The feasibility of proposed method includes watermark invisibility and the accuracy of extracted watermark. The embedding watermark should be invisible to human eyes in order to ensure the

Watermarked image					
PSNR	40.5680	40.4943	40.8568	40.7063	39.7095
Extracted watermark	BIT	BIT	BIT	BIT	BIT
NC	1	1	1	1	1
SSIM	1	1	1	1	1

Fig. 5 Embedding and extraction of watermark on different cover images

Method	rot=15	rot=30	rot=45	rot=60	rot=75	rot=90
LCWT+QR	BIT	BIT	BIT	BIT	BIT	BIT
DWT+QR						BIT
QR						BIT

Fig. 6

Fig. 6 Extracted watermark images under rotation attacks

information safety, so watermark invisibility is an important index to measure the power of method. Moreover, in order to guarantee the accuracy of extracted information, the accuracy of extracted watermark is another index we should consider. As follows, we use three metrics to measure the two aspects.

First, PSNR is commonly used as an index to measure the quality of reconstruction watermarked image, which is defined as

$$\text{PSNR} = 10 \log_{10} \frac{MN \cdot [\max(f(m, n))]^2}{\sum_{m=1}^M \sum_{n=1}^N [f(m, n) - f^w(m, n)]^2} \quad (17)$$

where f is original image and f^w is watermarked image with same size $M \times N$.

In addition, in order to measure the accuracy of extracted watermark, the NC and the SSIM [29] are used to compare the quality of extracted watermark and original watermark image. The NC is defined as

$$\text{NC} = \frac{\sum_{m=1}^M \sum_{n=1}^N w(m, n) \cdot w'(m, n)}{\sqrt{\sum_{m=1}^M \sum_{n=1}^N [w(m, n)]^2} \cdot \sqrt{\sum_{m=1}^M \sum_{n=1}^N [w'(m, n)]^2}} \quad (18)$$

where M and N are the size of original watermark image w and extracted watermark image w' .

The SSIM is based on the degradation of structural information, which is defined as follows:

$$\text{SSIM}(w, w') = \frac{(\mu_w \mu_{w'} + c_1)(2\sigma_{ww'} + c_2)}{(\mu_w^2 + \mu_{w'}^2 + c_1)(\sigma_w^2 + \sigma_{w'}^2 + c_2)} \quad (19)$$

where μ_w and $\mu_{w'}$ are average of images w and w' . σ_w and $\sigma_{w'}$ are variance of images w and w' . $\sigma_{ww'}$ is the covariance of images w and w' , and c_1 and c_2 are two variables to stabilise the division with a weak denominator.

Based on the consideration of the above facts, the watermark is embedded into the five cover images. The five watermarked cover images and corresponding PSNR values are listed in Fig. 5. It can be seen from Fig. 5 that there is no visual quality difference between watermarked cover images and original cover images shown in Fig. 4. Generally speaking, the human eye cannot distinguish the difference between watermarked cover images and original cover images when the PSNR is larger than 30 dB. The larger PSNR value we get, the better quality of watermarked cover image is. It can be found that the PSNR listed in Fig. 5 are larger than 30 dB. This implies that our method has better watermark invisibility.

Moreover, in order to evaluate the accuracy of extraction watermark, the extracted watermark images are compared with original watermark images by the NC and SSIM values. The extracted watermark images and the corresponding NC and SSIM values are listed in Fig. 6. The NC and SSIM values are equal to 1, which demonstrate that the proposed method can extract the watermark correctly. Based on the above results, we can derive the conclusion that the proposed method is feasible.

4.2 Robustness of proposed method

The watermarked cover image is often destroyed by various attacks, such as the image rotation, the image scaling, the image compression and the image filter. The robustness is a key factor to show the performance of a watermarking method. In this

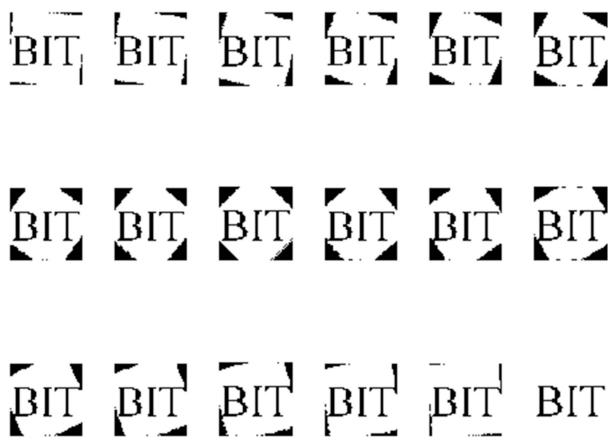


Fig. 7 Extracted watermark images under rotate attacks with 5° – 90°

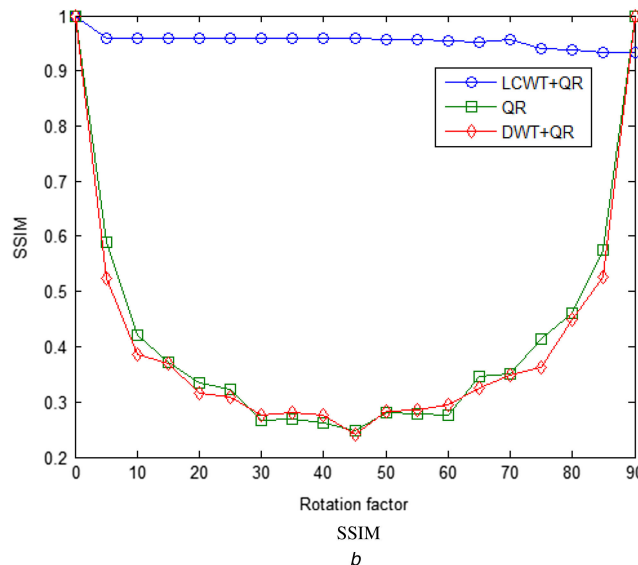
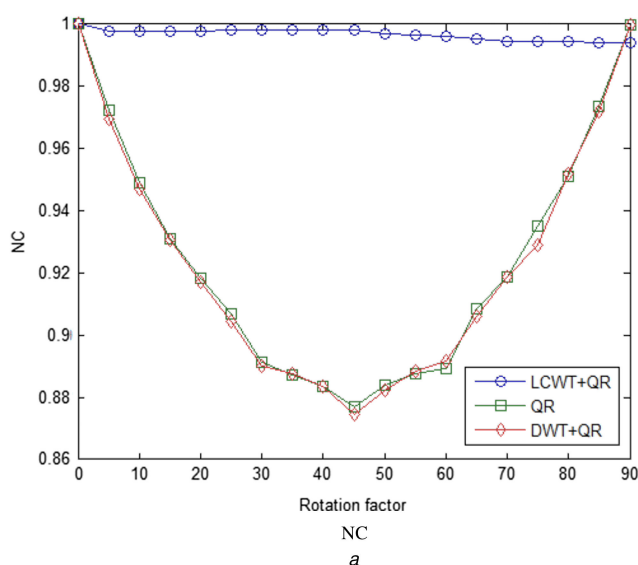


Fig. 8 Performance of three methods under rotation attacks (a) NC, (b) SSIM

subsection, the robustness of the proposed watermarking method are demonstrated by performing several attack operations on watermarked cover image and then extract the watermark from attacked image. These attack operations include JPEG compression, scaling, rotating, addition of noise and filter and so on. In the following, the proposed method is compared with the method proposed in [17, 18]. The Haar wavelet together with decomposition level of 1, are used in the DWT. In order to guarantee the fair of comparison, we assume that all parameters using in three methods are selected in order to obtain almost same PSNR value around 40 dB. So based on the above assumption, $\lambda = 15$ is used in the proposed method, watermark strength $S = 35$ is used in [17, 18].

4.2.1 Image rotation robustness: The image rotation robustness are explored when the watermarked cover image is attacked by the rotation with 5° – 90° . First, the extracted watermark image under rotation attacks is shown in Fig. 7. It can be seen from Fig. 7 that all these extracted watermark images have large errors (the black part around the 'BIT'). By further analysis, we find that these errors have a relationship with the rotation factor. So here we use a universal technology means to decrease these errors. If the distance of the element w_{ij} to centre point larger than 31, we set $w_{ij} = 1$, else w_{ij} equal to the original value. Then, we consider the relationship between the NC, SSIM values and the rotation factors range from 5° to 90° with interval step 5° , the results are shown in Figs. 8a and b, respectively. To make the results more intuitive, we present the

extracted watermark images in Fig. 6. Through analysis and comparison, we find that the NC and SSIM values of the proposed method are higher than the other two methods. From a subjective perception, the extracted watermark images using the proposed method are more clear than the other two methods. Based on the above results, we can obtain that the proposed method has a better robustness to rotation attack.

4.2.2 Image scaling robustness: It is necessary to detect the image scaling robustness when the watermarked cover image is attacked by scaling. We consider the relationship between the NC, SSIM values and the scaling factors range from 0.2 to 1.2 with interval step 0.2, the results are shown in Figs. 9a and b, respectively. Moreover, we also present the extracted watermark images in Fig. 10 in order to make the results more intuitive. Through analysis and comparison of NC and SSIM values based on three methods, we find that the NC and SSIM values are increased with the increasing of scaling factors. The proposed method jumps at scaling factor 0.4, so the proposed method can arrive the good performance quickly. From a subjective perception,

the extracted watermark images based on proposed method are more clear than the other two methods at scaling factors 0.4, 0.6, 0.8 and 1.2. Based on the above results, the proposed method is better than the other two methods under the case of scaling attack.

4.2.3 Image compression robustness: When image transmitted through the Internet, the image, usually compressed in order to speed up transmission rate and save transmission time. So it is necessary to detect the image JPEG compression robustness. We detect the relationship between the NC, SSIM values based on three methods and the JPEG compression ratio range from 5 to 90 with interval step 5, the results are shown in Figs. 11a and b, respectively. The extracted watermark images are listed in Fig. 12. Through analysis and comparison of NC and SSIM values based on the three methods, we can find that the NC and SSIM values are increased with the increasing of compression factors. Due to the algorithm of watermark embedding and extraction based on the proposed method is different from QR and DWT+QR, which cause the phenomenon that the proposed method is not better than QR or DWT+QR for the compression attack with the JPEG factor below 30. However, the proposed method has a jump at compression factor 30 in both the NC and SSIM values. In other words, the proposed method can arrive the good performance quickly. From Fig. 12, the extracted watermark images based on proposed method become more and more clear after compression factor 30, subjective perception are in agreement with the meaning of the data. The extracted watermark images are more clear than

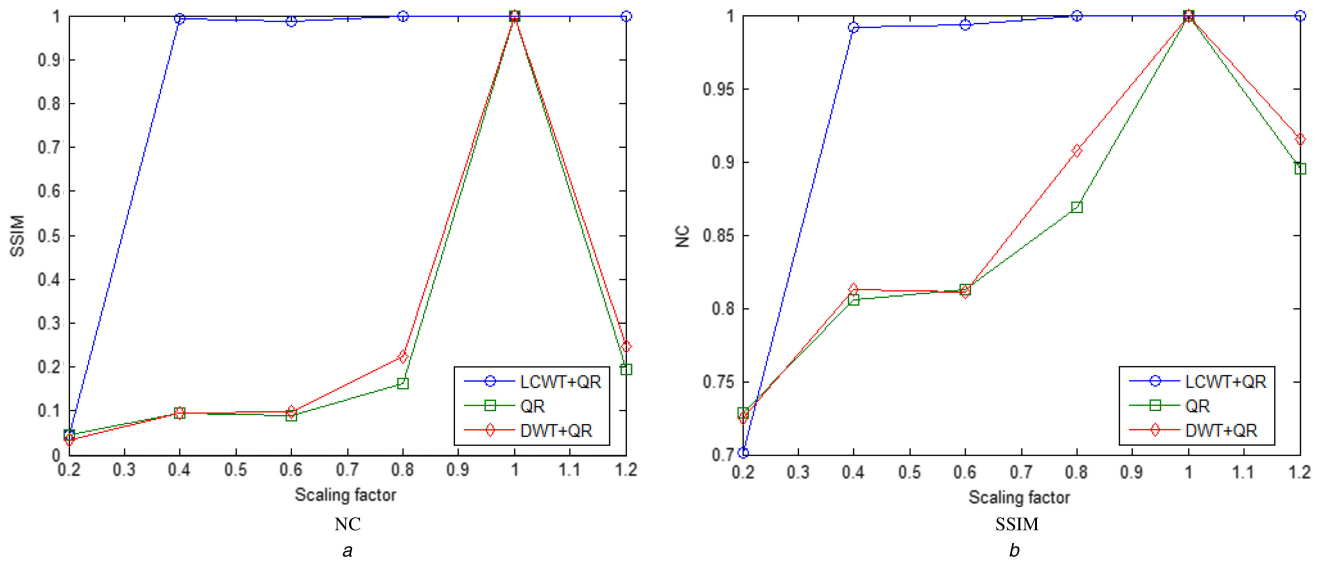


Fig. 9 Performance of three methods under scaling attacks
(a) NC, (b) SSIM

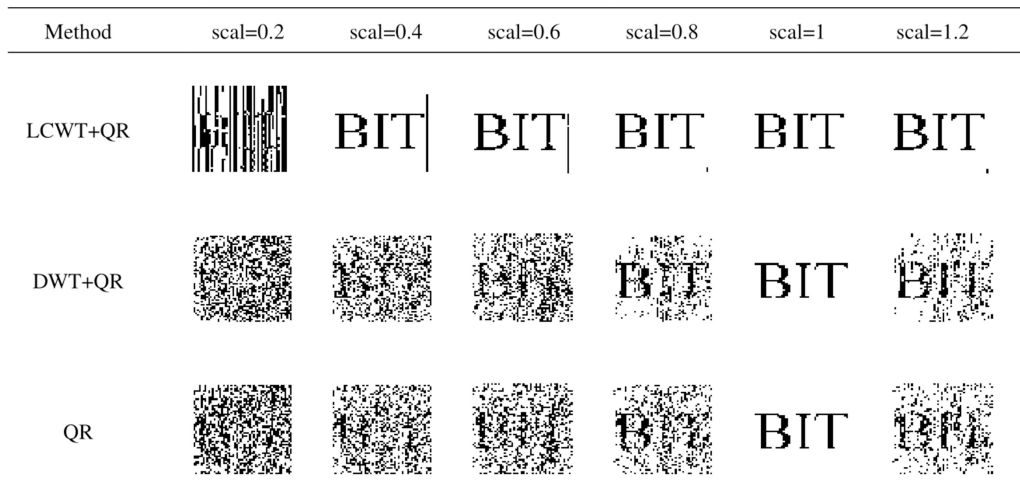


Fig. 10 Extracted watermark images under scaling attacks

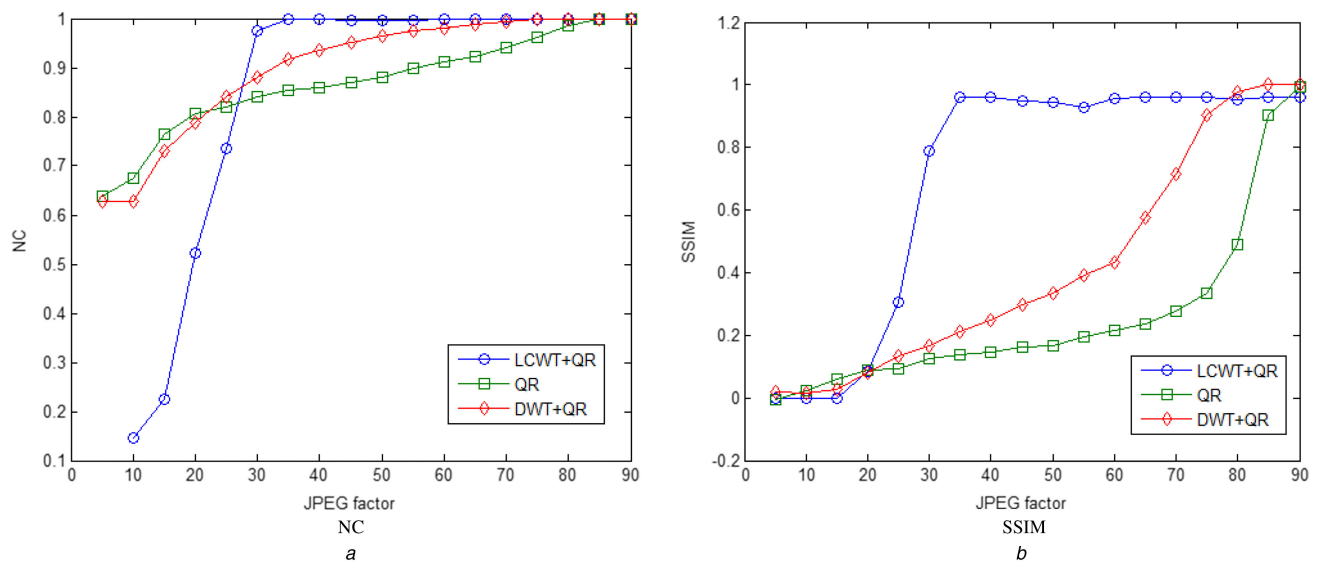


Fig. 11 Performance of three methods under JPEG compression attacks
(a) NC, (b) SSIM

the other two methods at compression factors 30, 45, 60, 75. In brief, compared with the watermark method based on DWT + QR

and QR, the proposed method is more robust to JPEG compression.

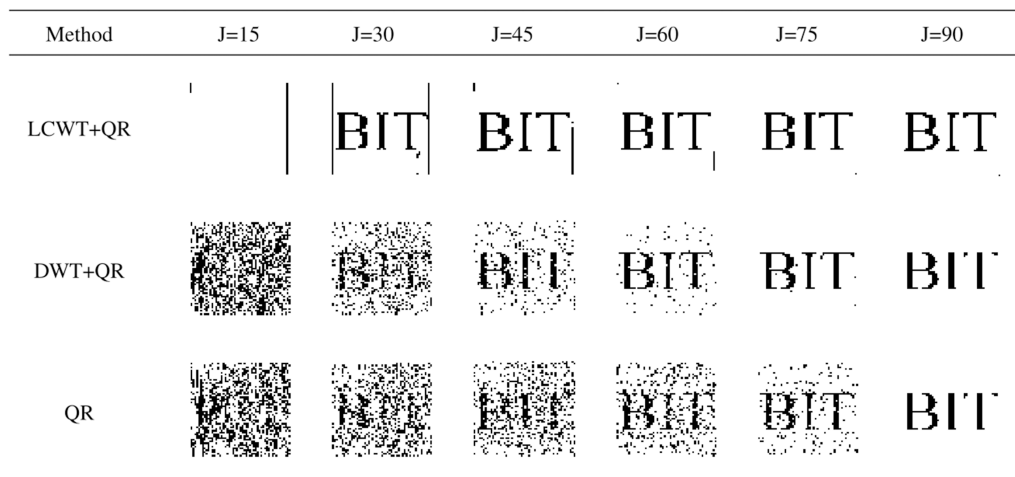


Fig. 12 Extracted watermark images under JPEG compression attacks

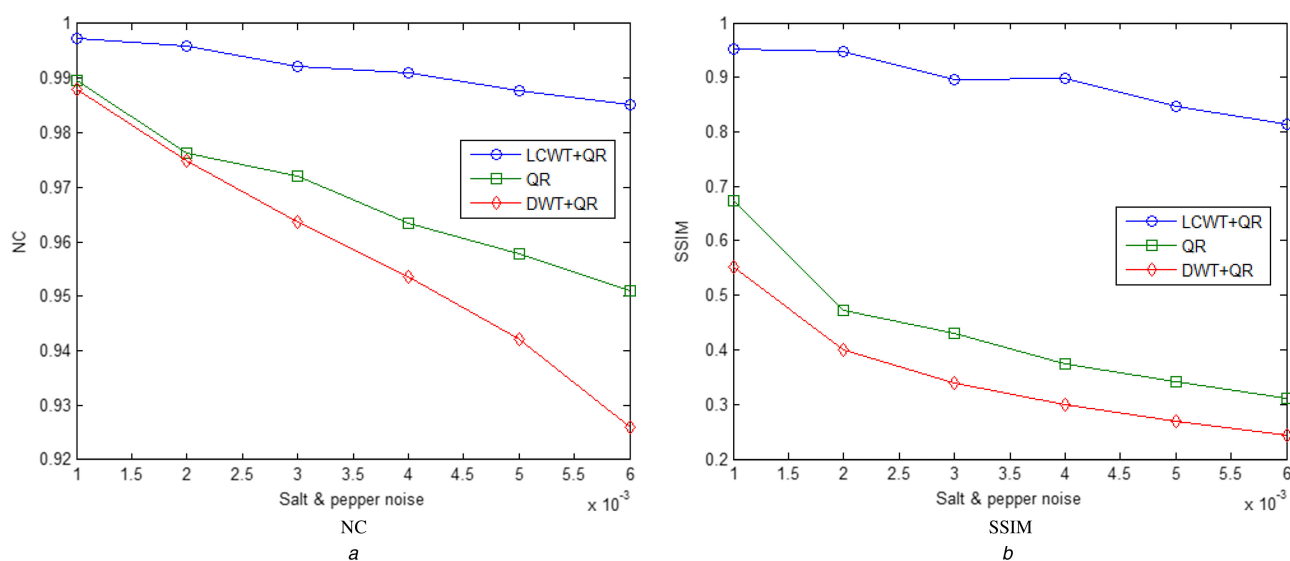


Fig. 13 Performance of three methods under salt & pepper noise attacks (a) NC, (b) SSIM

4.2.4 Image noise robustness: The transmission of image often polluted by different noises, so it is necessary to detect the image noise robustness when the watermarked cover image is polluted by different noise. In this part, three common noises, including Gaussian noise, salt & pepper noise and speckle noise are considered. This part is divided into three categories.

First, we consider the relationship between the NC, SSIM values and the noise density of salt & pepper range from 0.1 to 0.6% with interval step 0.1%, the results are shown in Figs. 13a and b, respectively. The extracted watermark images are listed in Fig. 14. Through analysis and comparison of the NC and SSIM values based on three methods, we find that the NC and SSIM values are decreasing with the increasing of noise density. However, the NC and SSIM values based on the proposed method are higher than the other two methods at all noise density. From a subjective perception, the extracted watermark images based on the proposed method are more clear than the other two methods at noise density 0.1–0.6%. At noise density 0.6%, the extracted watermark images based on proposed method is relatively clearer than the extracted watermark images based on other two methods. Based on the above results, the proposed method has a better robustness to salt & pepper noise attack.

Second, we detect the relationship between the NC, SSIM values and the variance of Gaussian noise range from 0.1 to 0.6% with interval step 0.1%, the results are shown in Figs. 15a and b, respectively. To make the results more intuitive, the extracted watermark images are presented in Fig. 16. Through analysis and comparison of the NC and SSIM values based on three methods,

the NC and SSIM values based on the proposed method are higher than the other two methods at all variance of Gaussian noise. From subjective perception, the extracted watermark images based on proposed method are relatively clear. In contrast to, the extracted watermark images based on other two methods are too blurry to distinguish. It means that the image watermarking method based on the proposed method has a better robustness to Gaussian noise attack than the other two methods.

Finally, we detect the relationship between the NC, SSIM values based on three methods and the variance of speckle noise range from 0.1 to 0.6% with interval step 0.1%, the results are shown in Figs. 17a and b, respectively. To make the results more intuitive, we present the extracted watermark images in Fig. 18. Through analysis and comparison of the NC and SSIM values based on three methods, the NC and SSIM values based on the proposed method decrease slowly, but the NC and SSIM values based on the other two methods decrease quickly. From a subjective perception, the clarity of extracted watermark images based on proposed method is decreasing slowly. In contrast to, the extracted watermark images based on other two methods are decreasing quickly. The extracted watermark is too blurry to distinguish after $\sigma = 0.3\%$. Based on the above facts, the proposed method is more robust against to speckle noise compared with the other two methods.

4.2.5 Image filter robustness: The poor transmission of image is often smoothed to decrease spurious effects, smoothing operation is usually by filter using median filter, Gaussian filter and average

Method	$\sigma = 0.1\%$	$\sigma = 0.2\%$	$\sigma = 0.3\%$	$\sigma = 0.4\%$	$\sigma = 0.5\%$	$\sigma = 0.6\%$
LCWT+QR						
DWT+QR						
QR						

Fig. 14 Extracted watermark images under salt & pepper noise attacks

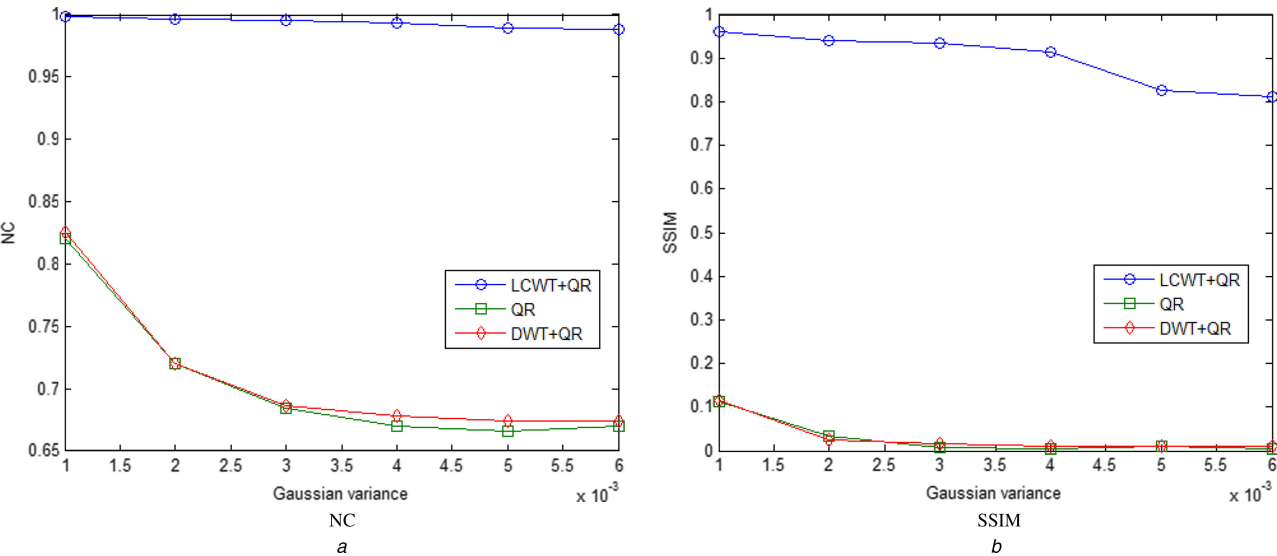


Fig. 15 Performance of three methods under Gaussian noise attacks (a) NC, (b) SSIM

Method	$\sigma = 0.1\%$	$\sigma = 0.2\%$	$\sigma = 0.3\%$	$\sigma = 0.4\%$	$\sigma = 0.5\%$	$\sigma = 0.6\%$
LCWT+QR						
DWT+QR						
QR						

Fig. 16 Extracted watermark images under Gaussian noise attacks

filter. It is necessary to detect the image filter robustness when the watermarked cover image is attacked by the filter. In this part, three common filters, including Gaussian filter, median filter and average filter are considered. This part is divided into two categories.

First, we consider the relationship between the NC, SSIM values based on three methods and the standard deviation of Gaussian lowpass filter range from 0.6 to 3.6 with interval step 0.6, the results are shown in Figs. 19a and b, respectively. Moreover, we also present the extracted watermark images in Fig. 20. Through analysis and comparison of the NC and SSIM values are

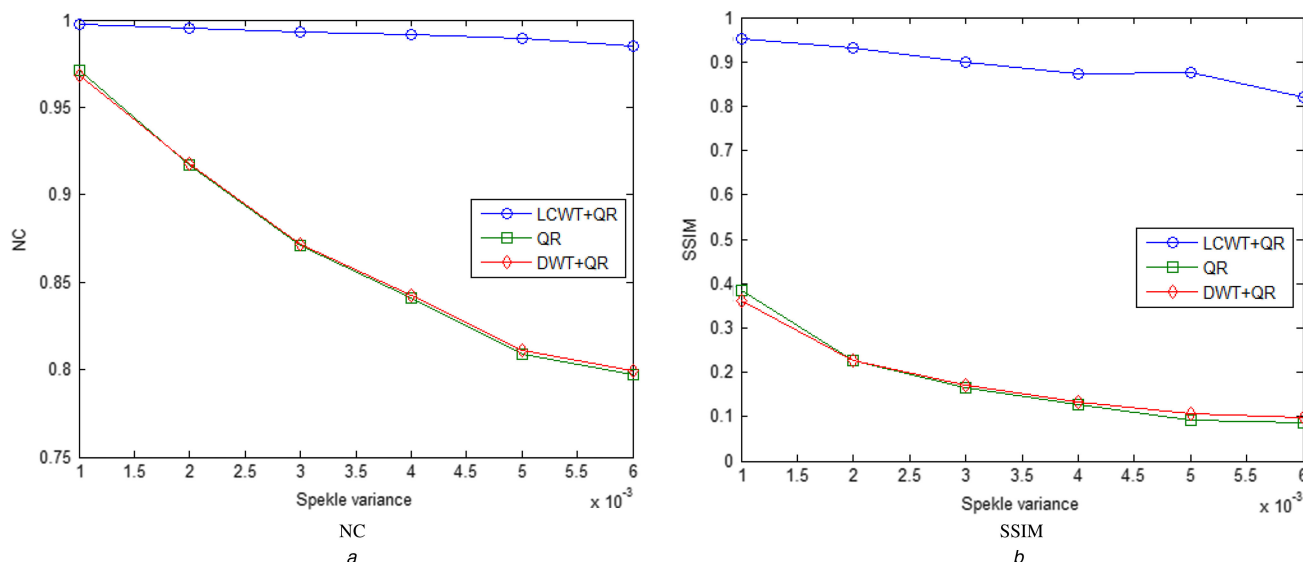


Fig. 17 Performance of three methods under speckle noise attacks
(a) NC, (b) SSIM

Method	$\sigma = 0.1\%$	$\sigma = 0.2\%$	$\sigma = 0.3\%$	$\sigma = 0.4\%$	$\sigma = 0.5\%$	$\sigma = 0.6\%$
LCWT+QR						
DWT+QR						
QR						

Fig. 18 Extracted watermark images under speckle noise attacks

listed in Figs. 19a and b, the NC and SSIM values are decreasing slowly with the increasing of standard deviation of Gaussian lowpass filter. However, the NC and SSIM values based on the proposed method are higher than the other two methods at all standard deviation. From a subjective perception, the extracted watermark images based on proposed method are more clear than the other two methods. At standard deviation $std=3.6$, the extracted watermark images based on proposed method are relatively clear, but the extracted watermark images based on other two methods are blurry. So we can draw the conclusion that the proposed method has a better robustness to Gaussian filter attack.

Second, the NC, SSIM values and the extracted watermark images under median filter and average filter attacks are shown in Fig. 21. The SSIM and NC values based on the proposed method are higher than the other two methods. By comparison of subjective, the clarity of extracted watermark based on the proposed method is best, which can also be obtained from the meaning of the NC and SSIM values.

4.2.6 Image translation and cropping robustness: Translation and cropping are two types of common geometrical attacks. It is necessary to detect the image translation and cropping robustness when the watermarked cover images are attacked by translation and cropping. In this part, we consider that the watermarked cover image is shifted with 16×16 pixels to the right and down and cropped one forth from upside. The extracted watermark images and the corresponding NC and SSIM values are presented in

Fig. 22. For translation, all three methods can extract clear watermark images, where the proposed method slightly better than the other two methods. For cropping, the proposed method can extract clear and entire watermark image. The extracted watermark images based on the other two methods are clear, but not entire. Summarising the above analysis, the proposed method provides better performance.

4.2.7 Some image processing attacks robustness: In this part, we will study the robustness of some image processing attacks including Sobel sharpen, motion blur with default value, histogram equalisation and gamma correction with 0.6. Histogram equalisation is used to enhance the overall image contrast, Sobel sharpen can enhance the image contour and details by Sobel edge detection algorithm. The extracted watermark images and the corresponding NC and SSIM values are shown in Fig. 23. From the analysis of Fig. 23, the proposed method outperforms the other two methods for all four image processing attacks. It means that the proposed method is robust against Sobel sharpen, motion blur, histogram equalisation and gamma correction to a certain extent.

5 Conclusions and future works

In this paper, a new transform named as linear canonical wavelet transform is introduced and then applied to digital image watermarking. First of all, inspired by the fact that WT can be written as a classical convolution form, so LCWT is introduced

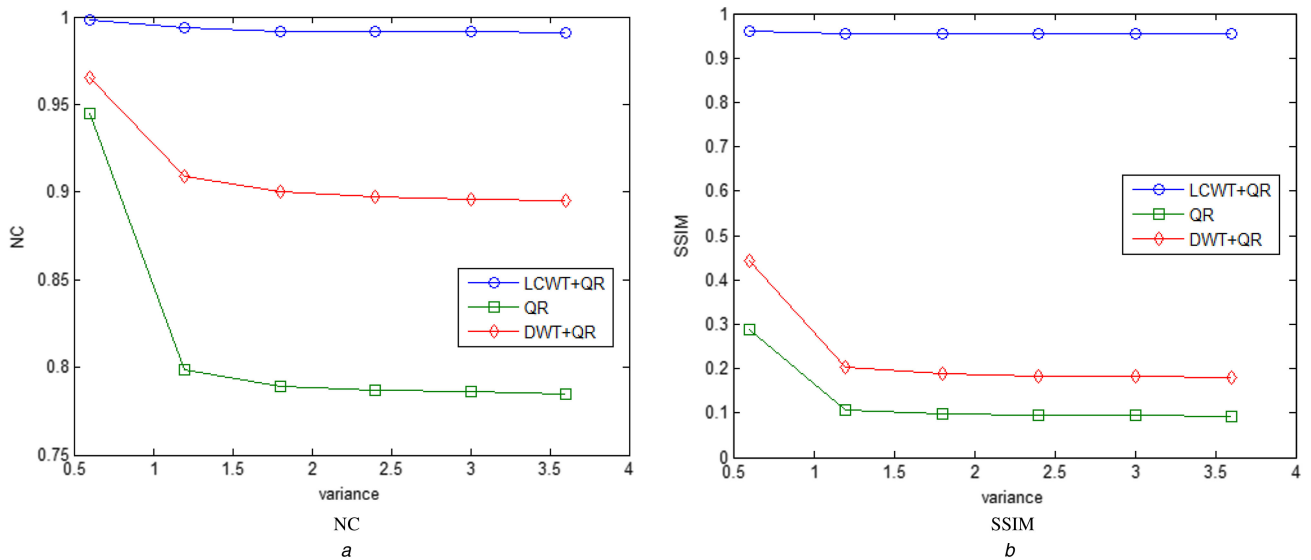


Fig. 19 Performance of three methods under Gaussian filter attacks (a) NC, (b) SSIM

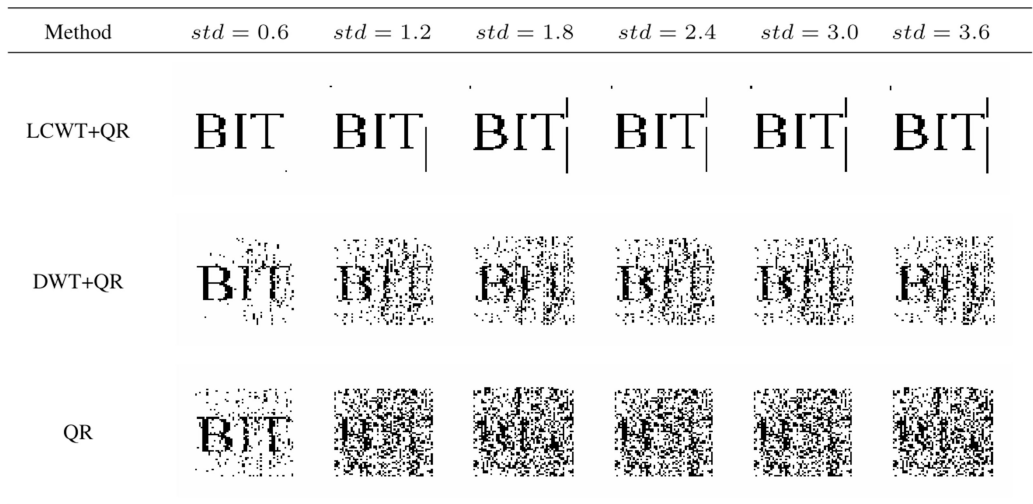


Fig. 20 Extracted watermark images under Gaussian filter attacks

Image operations	LCWT+QR	DWT+QR	QR
Median filter			
NC	0.9977	0.9356	0.9006
SSIM	0.9597	0.2528	0.1915
Average filter			
NC	0.9982	0.8936	0.7843
SSIM	0.9598	0.1795	0.0902

Fig. 21 Extracted watermark images under average and median filter attacks

based on general linear canonical convolution. It is worth to note that WT is the special case of LCWT. Moreover, a novel image watermarking method based on LCWT and QR decomposition is proposed. The feasibility and robustness of the image

watermarking method are detected. The experimental results are shown as follows:

- The LCWT can be applied to image watermarking successfully and the image watermarking method using LCWT can extract the watermark image correctly, which means that the proposed method is feasible in application of image watermarking.
- The image watermarking method based on LCWT and QR decomposition has better robustness to both geometric attacks (rotation, scaling, translation and cropping) and image processing attacks (addition of noise, JPEG compression and filter etc.). Compared with the watermark method proposed in [17, 18], the performance of the proposed method is better.

Our method can be further improved and extended to other fields. The effects of the parameters of LCWT on watermark invisibility and robustness are our further research focus.

6 Acknowledgments

This work was supported by the National Natural Science Foundation of China (no. 61171195), by Program for New Century Excellent Talents in University (No. NCET-12-0042), and also by the Foundation for Innovative Research Groups of the National Natural Science Foundation of China (no. 61421001).

Attack operations	LCWT+QR	DWT+QR	QR
Translation	BIT	BIT	BIT
NC	0.9996	0.9970	0.9970
SSIM	1	0.9753	0.9753
Cropping	BIT	BIT	BIT
NC	0.9943	0.9829	0.9829
SSIM	0.9956	0.8882	0.8882

Fig. 22 Extracted watermark images under translation and cropping attacks

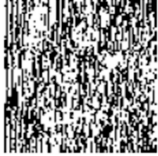
Attack operations	LCWT+QR	DWT+QR	QR
Sobel Sharpen			
NC	0.9151	0.6663	0.7325
SSIM	0.3469	0.0035	0.0154
Motion Blur			
NC	0.9410	0.7329	0.7390
SSIM	0.6003	0.0307	0.0432
Histogram Equalization			
NC	0.9912	0.6703	0.6701
SSIM	0.9965	0.0145	0.0131
Gamma Correction			
NC	0.9997	0.5540	0.7024
SSIM	0.9996	0.0389	0.0279

Fig. 23 Extracted watermark images under some image processing attacks

7 References

- [1] Lang, J., Zhang, Z.: 'Blind digital watermarking method in the fractional Fourier transform domain', *Opt. Lasers Eng.*, 2014, **53**, pp. 112–121
- [2] Qi, M., Li, B.Z., Sun, H.F.: 'Image watermarking via fractional polar harmonic transforms', *J. Electron. Imaging*, 2015, **24**, (1), pp. 1–12

- [3] Muhammad, N., Bibi, N.: 'Digital image watermarking using partial pivoting lower and upper triangular decomposition into the wavelet domain', *IET Image Process.*, 2015, **9**, (9), pp. 795–803
- [4] Singh, C., Ranade, S.K.: 'Image adaptive and high-capacity watermarking system using accurate Zernike moments', *IET Image Process.*, 2014, **8**, (7), pp. 373–382
- [5] Qi, M., Li, B.Z., Sun, H.F.: 'Image watermarking using polar harmonic transform with parameters in SL (2, R)', *Signal Process., Image Commun.*, 2015, **31**, pp. 161–173
- [6] Zhang, Q.Y., Xing, P.F., Huang, Y.B., *et al.*: 'An efficient speech perceptual hashing authentication algorithm based on wavelet packet decomposition', *J. Inf. Hiding Multimedia Signal Process.*, 2015, **6**, (2), pp. 311–322
- [7] Okada, M., Okabe, Y., Uehara, T.: 'A web-based privacy-secure content trading system for small content providers using semi-blind digital watermarking'. Seventh IEEE Consumer Communications and Networking Conf. (CCNC), 2010, 2010, pp. 1–2
- [8] Mehta, R., Rajpal, N., Vishwakarma, V.P.: 'LWT-QR decomposition based robust and efficient image watermarking scheme using Lagrangian SVR', *Multimedia Tools Appl.*, 2015, **75**, pp. 1–22
- [9] Ansari, I.A., Pant, M., Ahn, C.W.: 'Robust and false positive free watermarking in IWT domain using SVD and ABC', *Eng. Appl. Artif. Intell.*, 2016, **49**, pp. 114–125
- [10] Makbol, N.M., Khoo, B.E.: 'Robust blind image watermarking scheme based on redundant discrete wavelet transform and singular value decomposition', *AEU-Int. J. Electron. Commun.*, 2013, **67**, (2), pp. 102–112
- [11] Barni, M., Bartolini, F., Piva, A.: 'Improved wavelet-based watermarking through pixel-wise masking', *IEEE Trans. Image Process.*, 2001, **10**, (5), pp. 783–791
- [12] Lin, C.C., Chang, C.C., Chen, Y.H.: 'A novel SVD-based watermarking scheme for protecting rightful ownership of digital images', *J. Inf. Hiding Multimedia Signal Process.*, 2014, **5**, (2), pp. 124–143
- [13] Bao, P., Ma, X.: 'Image adaptive watermarking using wavelet domain singular value decomposition', *IEEE Trans. Circuits Syst. Video Technol.*, 2005, **15**, (1), pp. 96–102
- [14] Ye, X., Chen, X., Deng, M., *et al.*: 'A SIFT-based DWT-SVD blind watermark method against geometrical attacks'. Seventh Int. Congress on Image and Signal Processing (CISP), 2014, 2014, pp. 323–329
- [15] Su, Q., Niu, Y., Wang, G., *et al.*: 'Color image blind watermarking scheme based on QR decomposition', *Signal Process.*, 2014, **94**, pp. 219–235
- [16] Mohammad, A.A.: 'A new digital image watermarking scheme based on Schur decomposition', *Multimedia Tools Appl.*, 2012, **59**, (3), pp. 851–883
- [17] Naderahmadian, Y., Hosseini-Khayat, S.: 'Fast and robust watermarking in still images based on QR decomposition', *Multimedia Tools Appl.*, 2014, **72**, (3), pp. 2597–2618
- [18] Naderahmadian, Y., Hosseini-Khayat, S.: 'Fast watermarking based on QR decomposition in Wavelet domain'. Sixth Int. Conf. on Intelligent Information Hiding and Multimedia Signal Processing (IIH-MSP), 2010, 2010, pp. 127–130
- [19] Wang, S., Zhao, W., Wang, Z.: 'A watermarking algorithm based on QR-decomposition in DWT domain', *Microelectron. Comput.*, 2008, **25**, (5), pp. 51–54
- [20] Ali, M., Ahn, C.W.: 'An optimized watermarking technique based on self-adaptive DE in DWT-CSVD transform domain', *Signal Process.*, 2014, **94**, pp. 545–556
- [21] Koc, A., Ozaktas, H.M., Candan, C., *et al.*: 'Digital computation of linear canonical transforms', *IEEE Trans. Signal Process.*, 2008, **56**, (6), pp. 2383–2394
- [22] Healy, J.J., Sheridan, J.T.: 'Sampling and discretization of the linear canonical transform', *Signal Process.*, 2009, **89**, (4), pp. 641–648
- [23] Li, Y.G., Li, B.Z., Sun, H.F.: 'Uncertainty principles for Wigner-Ville distribution associated with the linear canonical transforms', *Abstr. Appl. Anal.*, 2014, **2014**, pp. 1–9
- [24] Shi, J., Liu, X., Zhang, N.T.: 'Generalized convolution and product theorems associated with linear canonical transform', *Signal Image Video Process.*, 2014, **8**, (5), pp. 967–974
- [25] Shi, J., Sha, X., Zhang, Q., *et al.*: 'Extrapolation of bandlimited signals in linear canonical transform domain', *IEEE Trans. Signal Process.*, 2012, **60**, (3), pp. 1502–1508
- [26] Xu, T.Z., Li, B.Z.: 'Linear canonical transform and its application' (Science Press, Beijing, 2013)
- [27] Wei, D., Li, Y.M.: 'Generalized wavelet transform based on the convolution operator in the linear canonical transform domain', *Optik-Int. J. Light Electron Opt.*, 2014, **125**, (16), pp. 4491–4496
- [28] Shi, J., Zhang, N.T., Liu, X.P.: 'A novel fractional wavelet transform and its applications', *Sci. Chin. Inf. Sci.*, 2012, **55**, (6), pp. 1270–1279
- [29] Wang, Z., Bovik, A.C., Sheikh, H.R., *et al.*: 'Image quality assessment: from error visibility to structural similarity', *IEEE Trans. Image Process.*, 2004, **13**, (4), pp. 600–612